Module IV
SUPERCONDUCTIVITY

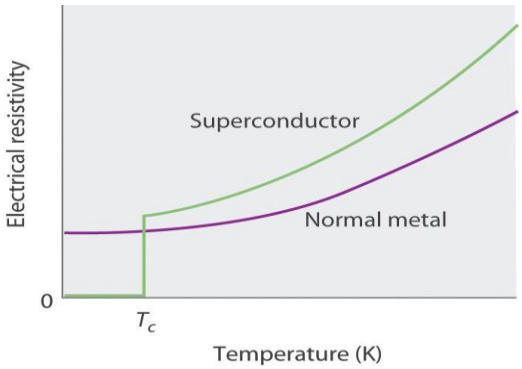
Introduction to Super conductors

A superconductor is a material that can conduct electricity without any resistance when it is cooled below a certain temperature known as the critical temperature (Tc). At this temperature, the material undergoes a phase transition that allows electric current to flow through it without energy loss. Superconductivity is a fascinating and unique quantum mechanical phenomenon, first discovered by Kamerlingh Onnes in 1911 when he observed that mercury's electrical resistance dropped to zero at very low temperature 4.15K.

A superconducting transition is a phase change in certain materials where they lose all electrical resistance and expel magnetic fields when cooled below a specific "critical temperature" (T_c).

Temperature Dependence of Resistivity in Superconducting Materials

In superconducting materials, the resistivity decreases with decrease in temperature and suddenly drops to zero at a particular temperature known as **critical temperature** (T_c) or **Transition temperature** (T_c) and remains zero below that as shown in Fig. The critical temperature for mercury is 4.2 K. Below this critical temperature. Mercury is superconducting whereas above this temperature it behaves like a normal conductor. Different superconducting materials have different critical temperatures but the nature of the variation in resistivity with temperature remains more or less similar. Critical temperatures of some materials are,



Ex: Mercury – 4.2K, Gallium – 1.15K, Aluminum – 1.20K, Lead 7.15K.

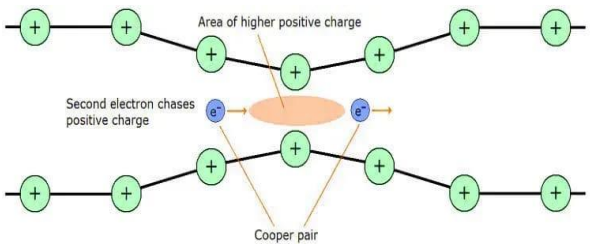
"The temperature at which resistance disappear is called **critical temperature**."

BCS Theory:

In 1957, Bardeen, Cooper and Schrieffer gave a theory to explain the phenomena of superconductivity, which is known as BCS Theory. This BCS theory is based upon the formation of cooper pairs, which is purely a quantum mechanical concept.

"When the electrons flow in the form of Cooper pairs in materials, they do not encounter any scattering and the resistance factor vanishes or, in other words, conductivity becomes infinity which is named as superconductivity. This is the essence of BCS theory"

When a free electron passes near an ion in a lattice, there is coulomb attraction between them. This causes deformation of the lattice. A second free electron passing nearby, takes advantage of the deformation to lower its energy. Thus the second electron interacts attractively with the first through the lattice deformation. This interaction is



called electron-lattice-electron interaction which can be considered to take place through phonons. The quantum of lattice vibrational energy is called phonon, similar to quantum of electromagnetic which energy which is called photon. The first electron emits a phonon which is absorbed by the second electron. Electrons in such an interaction have same energy and equal and opposite spins. Such pairs of electrons coupled together through a phonon are called Cooper pairs.

The attraction between electrons in a Cooper pair is very weak and the pair can be separated by a small increase in temperature which causes thermal vibrations.

At very low temperature, density of Cooper pairs is large and they move collectively through the lattice which minimizes collisions. Also, due to large density of Cooper pairs, their velocity is small even for large currents. Due to small velocity and the collective movement of the Cooper pairs, collisions are reduced and resistivity decreases.

The binding energy of cooper pair is called **Energy gap.** It is given by

$$E_g = 3.53kT_c$$

Where k is Boltzmann constant and T_c is critical temperature.

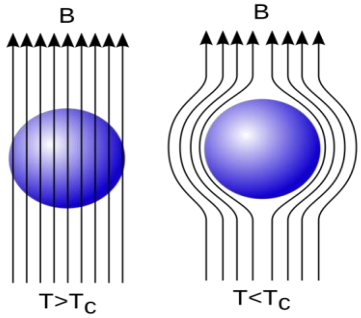
Persistent current:

A persistent current is a continuous flow of electric current that continues without an external power source, most notably found in superconductors due to their zero resistance.

When a current is setup in a closed loop of superconducting wire, this will continue to flow around the loop without any loss as long as the wire is kept below the critical temperature. Persistent currents in superconductors are often induced by a magnetic field and remain flowing after the field is removed, a phenomenon that is useful in applications like superconducting magnets. Experimental evidence shows that the decay time exceeds 10^5 years.

Meissner effect:

When a specimen is placed in a weak magnetic field and cooled below the critical temperature, the magnetic flux originally present in the specimen is ejected from the specimen. This effect is known as Meissner effect.



This effect is reversible and the specimen returns to its normal state when its temperature is raised above its critical temperature. Thus superconductors act as perfect diamagnets with zero magnetic induction in the interior when placed in weak magnetic fields.

The magnetic induction inside the specimen in the normal state is given by,

$$B=\mu_o (H + M)$$

Where H is the externally applied magnetic field and M is the magnetization inside the specimen.

When, temperature T of the specimen is lowered below its critical temperature Tc.

$$B= 0$$

$$\therefore \mu_o (H + M) = 0$$

$$H = - M$$

The susceptibility is given by, $\chi = \frac{M}{H} = -1$

This indicates perfect diamagnetism.

Since diamagnetic materials have negative magnetic susceptibility, the specimen becomes an ideal diamagnetic in superconducting state. If a specimen of superconductor is placed in a strong magnetic field, the specimen loses its property of superconductivity and becomes normal material.

Critical current:

The maximum current carried by a superconductor at a temperature below critical temperature without losing its superconductivity property is known as critical current(I_c).

A current carrying conductor produces magnetic field. The magnitude of magnetic field is directly proportional to strength of the current. If current in a superconductor exceeds critical current, then it may produce magnetic field which is stronger enough to destroy the superconducting property. Hence certain limitation on current flowing in a superconductor may be imposed. This is known as **Silsbee Rule.**

If I_c is the critical current and H_c is the critical field then for a cylindrical superconducting wire is

$$I_c = 2\pi R H_c$$

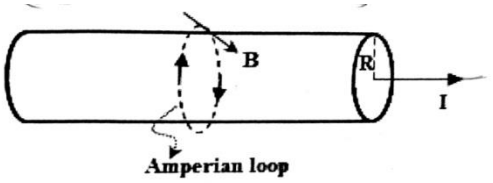
Critical current density J_c is given by

$$J_c = \frac{I_c}{a} = \frac{2\pi R H_c}{\pi R^2}$$

$$J_c = \frac{2H_c}{\pi R^2}$$

Derivation for a cylindrical wire using ampere's law

Consider a cylindrical wire of length L and radius R carrying current I. the magnetic field produced by the current on the surface of the cylinder is B. Imagine an Amperian loop of radius R around the cylinder as shown in fig



According to Ampere's law

$$\oint B.dl = \mu_o I \text{ -------- (1)}$$

The angle between the direction of B and the small elemental length dl is 0. Hence

$$B.dl = Bdl\cos 0 = Bdl$$

Equation (1) becomes

$$\oint Bdl = \mu_o I$$

$$B\oint dl = \mu_o I \text{ -------- (2)}$$

But $\oint dl = 2\pi R$ = circumference of the amperian loop

$$B(2\pi R) = \mu_o I$$

Substituting $B = \mu_o H$

$$\mu_o H(2\pi R) = \mu_o I$$

$$H = \frac{I}{2\pi R}$$

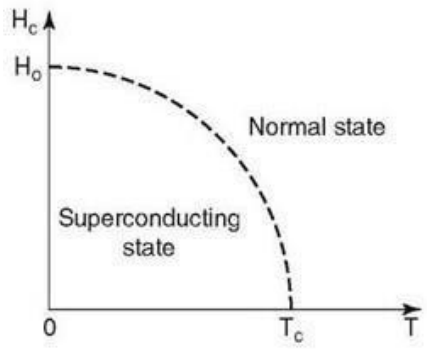
This is the magnetic field intensity. If the current is critical then

$$H = \frac{I_c}{2\pi R}$$

Effect of magnetic field on superconductors:

A superconducting material loses its resistivity at temperature T_c and becomes a superconductor. But at this stage, if it is subjected to a magnetic field, it becomes a normal conductor again. Then it needs to cool the material to still lower Temperatures for it to recover its superconducting property. However, it is possible to turn a superconductor into a normal conductor by subjecting it to a sufficiently strong magnetic field at all temperatures T < Tc. With this in view, we define a critical field as follows,

"The minimum value of the applied magnetic field required to destroy superconductivity is called the **critical field (H_c)"** and is a function of temperature. Above the critical temperature, $H_c = 0$. Below critical temperature the variation of H_c with temperature T is as shown in Fig.



For a superconducting material the critical field will be higher when temperature is lower. If T is the temperature of the superconducting material ($T < T_C$), H_C is the critical fieldand H_O is the field required to turn the superconductor to normal conductor at 0K.

Then the relation for critical field (H_C) is given by

$$H_c = H_o\left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

M-H Characteristics

Types of superconductors

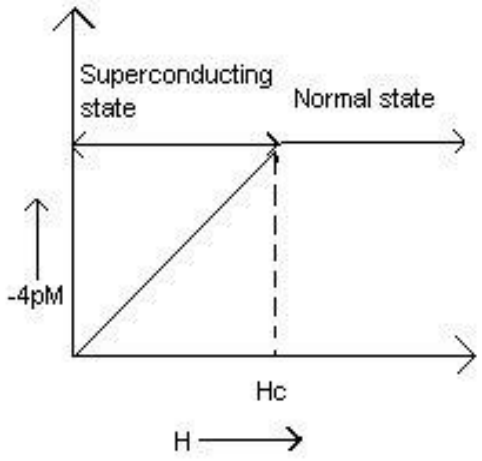
There are two types of superconductors based on the difference in magnetization exhibited by them. These are known as Type-I (soft) and Type-II (hard) superconductors.

Type-l Superconductors (Soft superconductors)

The magnetization curve for a Type-I superconductor is shown in Fig. Type-I superconductors show complete Meissner effect when the applied magnetic field H is less than the critical field.

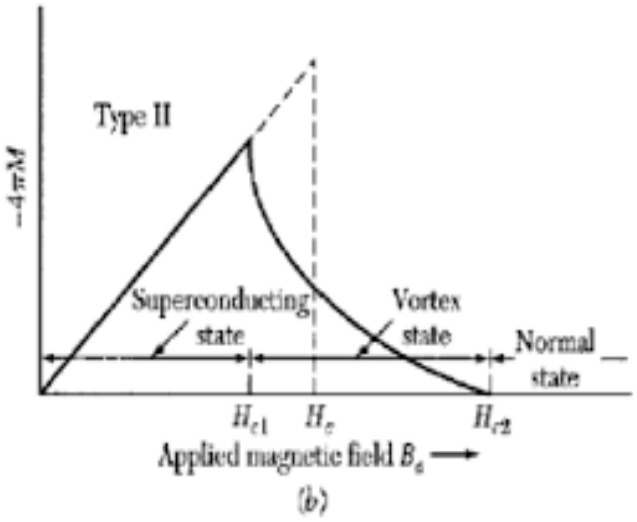
The magnetization in the specimen is proportional to the applied field and the specimen acts like a perfect diamagnet.

For $H>H_c$, the magnetization is negligible and the specimen becomes a normal conductor. Pure elements show such magnetization curves. The values of H_c are very low.



Type-II Superconductors (Hard superconductors)

Type-II superconductors are characterized by two critical fields Hc_1 and Hc_2. Up to the lower critical field Hc_1, the specimen is perfectly diamagnetic and flux is completely ejected out from the specimen. The magnetization in the specimen is proportional to the applied field. Between Hc_1 and Hc_2, Meissner effect is incomplete but electrically the specimen is a superconductor. This state is called the mixed state or the vortex state as the specimen is an electrically a superconductor but not magnetically.



The value of Hc_2 may be 100 times or more high than the critical field Hc for Type-I superconductors.

Above the higher critical field Hc2, the specimen becomes a normal conductor. Type-II superconductors are usually alloys or transition metals with high values of electrical resistivity in the normal state.

The materials which behave as Type-II are inhomogeneous and include Nb - Zn, Nb - Ti alloys and Va-Ga-and Nb - Sn intermetallic compounds.

Comparison between Type-I and Type II Superconductors

Sl.No.	Type-I Superconductor	Type-II Superconductor
1.	The material loses magnetization suddenly	The material loses magnetization gradually.
2.	They exhibit complete Meissner effect. i.e. they are completely diamagnetic.	They do not exhibit complete Meissner effect.
3.	They have only one critical magnetic field(H_c)	They have two critical magnetic field i.e. lower critical field(Hc_1) and upper critical field(Hc_2)
4.	No mixed states exists	Mixed state is present.
5.	Highest known critical field is 0.1 tesla	Critical field is much greater, i.e. upto 30 tesla
6.	They are also called soft superconductors as they lose their superconducting property at low value of magnetic field.	They are also called hard superconductors because they have a large amount of magnetic hysteresis.
7.	Ex: Lead, Tin and Mercury etc.,	Ex: Nb-Sn, Nb-Zn, Nb-Ti and Va-Ga, etc.,

High Temperature Superconductors

Kamerlingh Onnes discovered superconductivity in mercury in 1911 with a critical temperature of 4.2 K. Research was then focused on materials with larger critical temperatures so that the phenomenon could become commercially viable. But, till about 1986, the highest critical temperatures were observed to be nearly 20 K. In 1986 Georg Bednorz and Muller discovered that an oxide of lanthanum, barium and copper was observed to have a critical temperature of 36 K. In 1987, an oxide of yttrium, barium and copper was found to have a critical temperature of 90 K and then an oxide of thallium, barium, calcium and copper was found to have a critical temperature of 120 K. The main feature of these high temperature superconductors is the presence of parallel sheets of CuO_2. Efforts are going on to find superconductors with critical temperatures near room temperature.

There are some different crystal structures that result in high temperature superconductivity, all composed of layers of copper-oxygen structures sandwiched around layers of other elements. These are as follows

(i) Yttrium barium copper oxide ($YBa_2Cu_30_7$) with 92 K as Tc.

(ii) Bismuth strontium calcium copper oxide {$(BiPb)_2$ $Sr_2Ca_2Cu_3O_x$} with 105 K as Tc

(iii) Thallium barium calcium copper oxide ($TIBa_2Ca_2Cu_3O_{10}$) with 115 Kas Tc

(iv) Mercury barium calcium copper oxide ($HgBa_2$ Ca_2 Cu_3O_6) with 135 K as Tc

In view of their diverse applications, there has been a tremendous effort to find still more of them with even high TcThe hi-T_c, materials are all Type-II superconductors and have extremely high upper critical fields. However, new high temperature superconductors have a drawback as they are very brittle and do not carry enough current. Moreover, in the new hi-T_c superconductors, the formation of electrons pairs is not caused by the same electron lattice interaction as in the BCS theory. So far it not well understood that what does cause the formation of pairs. Intense research is being conducted in this direction.